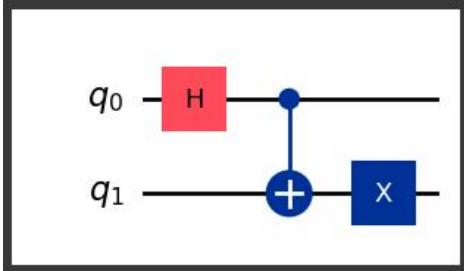
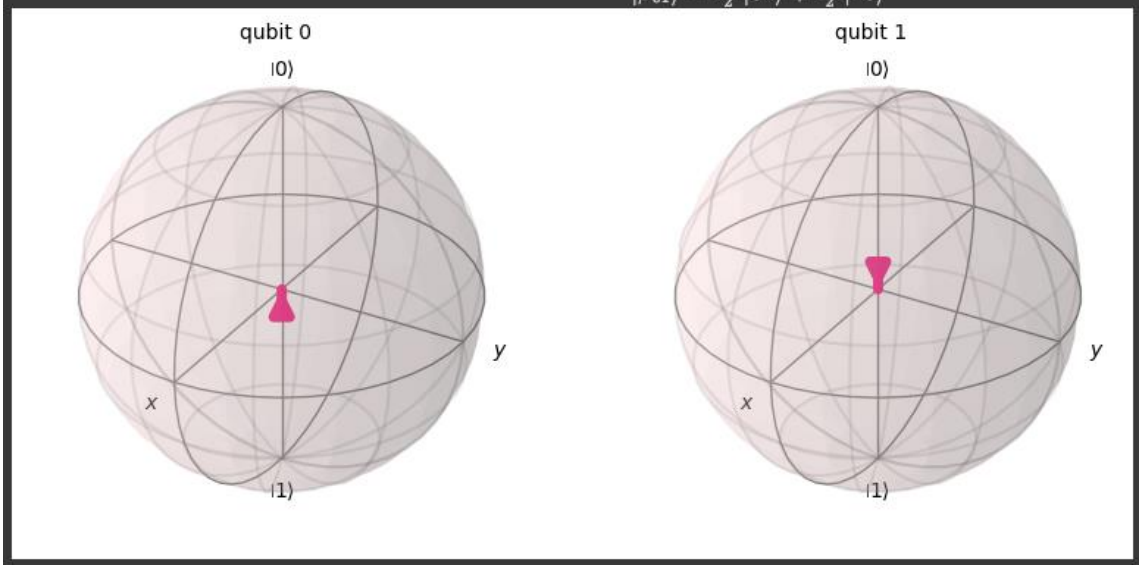
EJERCICIO ENTREGABLE 1: CREACIÓN DE ESTADOS DE BELL

Obtén circuitos para los estados de Bell |β01⟩, | β10⟩, | β11⟩, siguiendo las instrucciones disponibles en el notebook Puertas cuánticas:

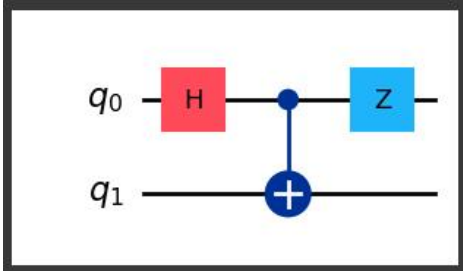
```
# |β_01⟩=(I⊗X¹Z⁰)|β_00⟩ = (I⊗X)|β_00⟩
qc = QuantumCircuit(2)

# Aplica las puertas para obtener |β_00⟩
qc.h(0)
qc.cx(0,1)

# Aplica la puerta X al cúbit que corrresponda
qc.x(1)

display(qc.draw('mpl'))

obten_estado(qc, "|\\beta_{01}\\rangle = ")
```



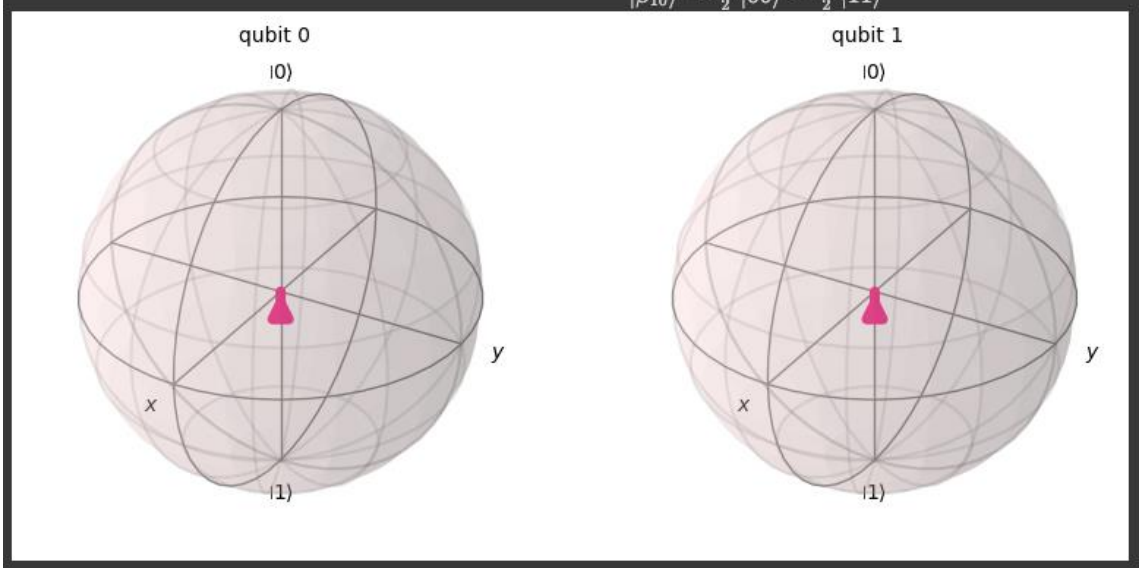
$|\beta_{01}\rangle = \frac{\sqrt{2}}{2}|01\rangle + \frac{\sqrt{2}}{2}|10\rangle$



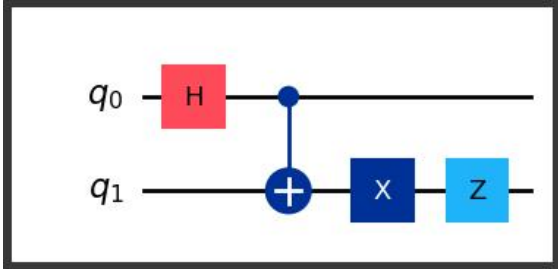
```python
# |β_01⟩=(I⊗X⁰Z¹)|β_00⟩ = (I⊗Z)|β_00⟩
qc = QuantumCircuit(2)

# Aplica las puertas para obtener |β_00⟩
qc.h(0)
qc.cx(0, 1)

# Aplica la otra puerta
qc.z(0)

display(qc.draw('mpl'))

obten_estado(qc, "|\\beta_{10}\\rangle = ")
```



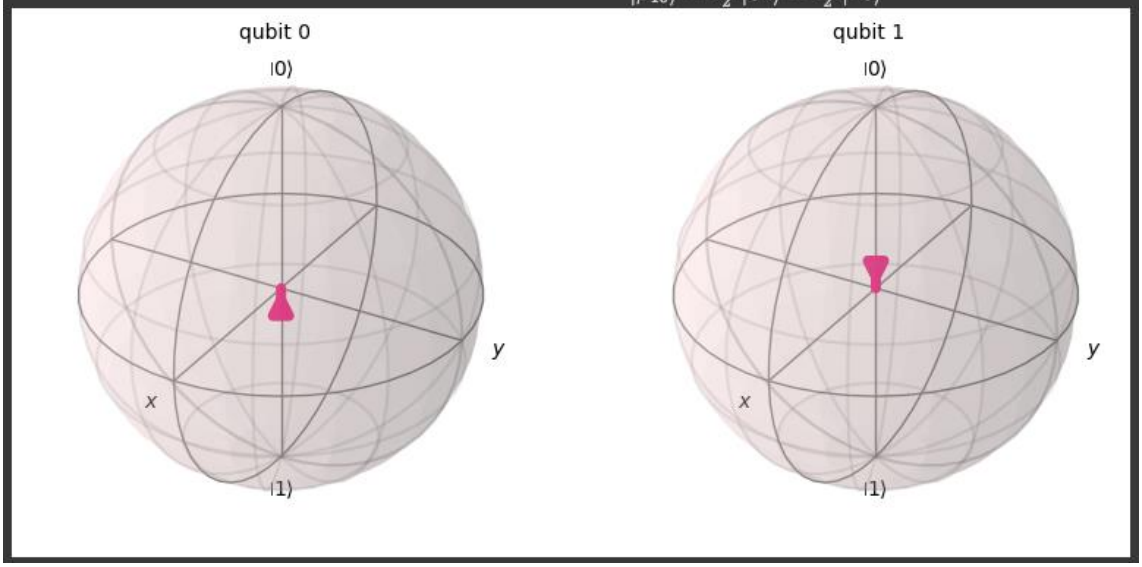
$|\beta_{10}\rangle = \frac{\sqrt{2}}{2}|00\rangle - \frac{\sqrt{2}}{2}|11\rangle$



```python
# |β_01⟩=(I⊗X¹Z¹)|β_00⟩ = (I⊗XZ)|β_00⟩
qc = QuantumCircuit(2)


# Aplica las puertas para obtener |β_00⟩
qc.h(0)
qc.cx(0, 1)

# Aplica las otras puertas
qc.x(1)
qc.z(1)

display(qc.draw('mpl'))

obten_estado(qc, "|\\beta_{10}\\rangle = ")
```



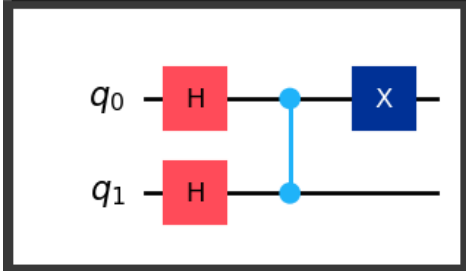
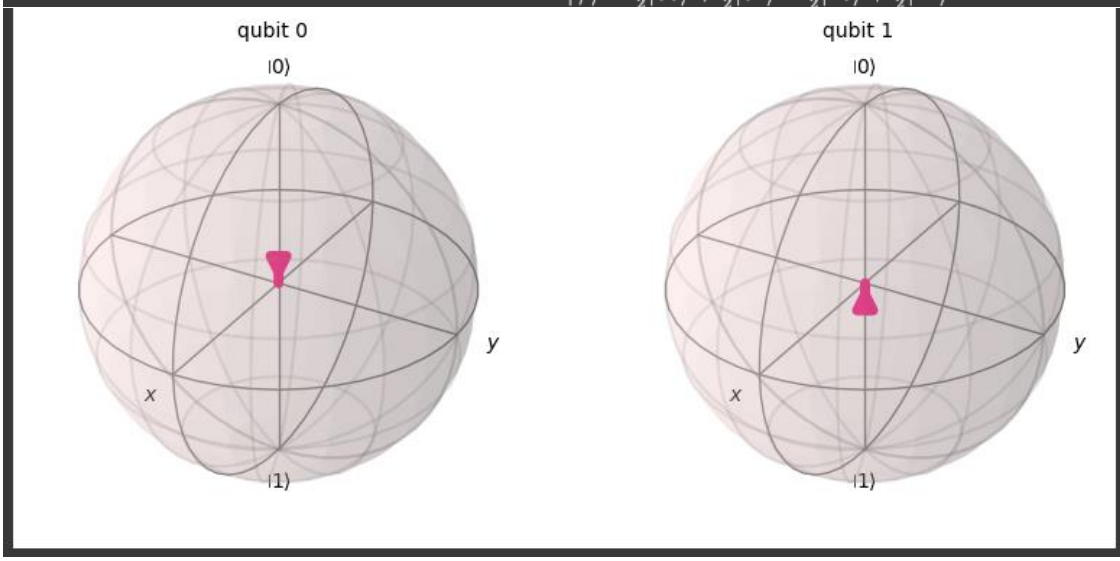
$|\beta_{10}\rangle = \frac{\sqrt{2}}{2}|01\rangle - \frac{\sqrt{2}}{2}|10\rangle$



EJERCICIO ENTREGABLE 2: Controlled-Z

Crea un circuito que obtenga el siguiente estado: $|\psi\rangle=1/2(|00\rangle+|01\rangle-|10\rangle+|11\rangle)$, siguiendo las instrucciones disponibles en el notebook Puertas cuánticas:

```
n = 2
qc = QuantumCircuit(n)

# Pon los cúbits en superposición
# para obtener el estado |ψ)=1/2(|00)+|01)+|10)+|11))
qc.h(0)
qc.h(1)

# La puerta C-Z cambia el signo del estado |11>
# Aplicamos la puerta C-Z para obtener
# el estado |ψ)=1/2(|00)+|01)+|10)-|11))
qc.cz(0,1)

# Aplicamos una puerta a q0 y para obtener
# |ψ)=1/2(|01)+|00)+|11)-|10))
qc.x(0)


display(qc.draw('mpl'))

obten_estado(qc, "|\psi\\rangle = ")
```



$|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$